02460: Advanced Machine Learning

Søren Hauberg

Week 6!

Metrics and geodesics

Recap of last week

We consider latent variable models on the form $\mathbf{y} = f(\mathbf{x})$

observation

latent representation

mapping: $f : \mathbb{R}^d \to \mathbb{R}^D, \qquad d \leq D$
(either deterministic of stochastic)
We call this the 'decoder'

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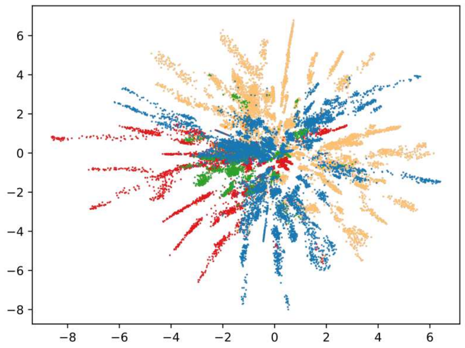
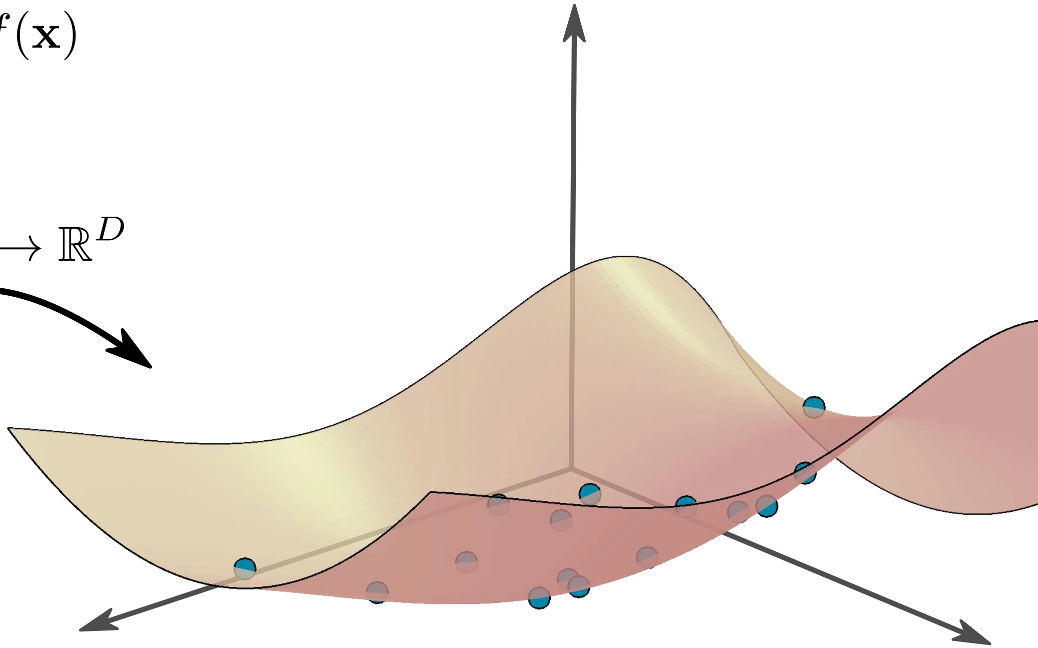


$f : \mathbb{R}^d \to \mathbb{R}^D$



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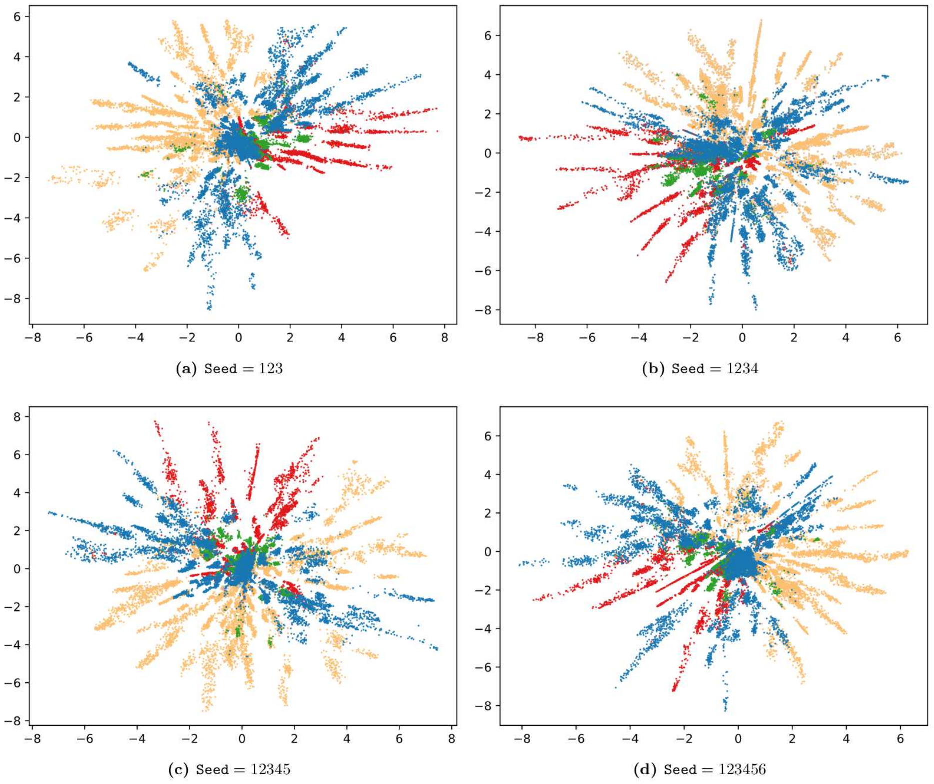
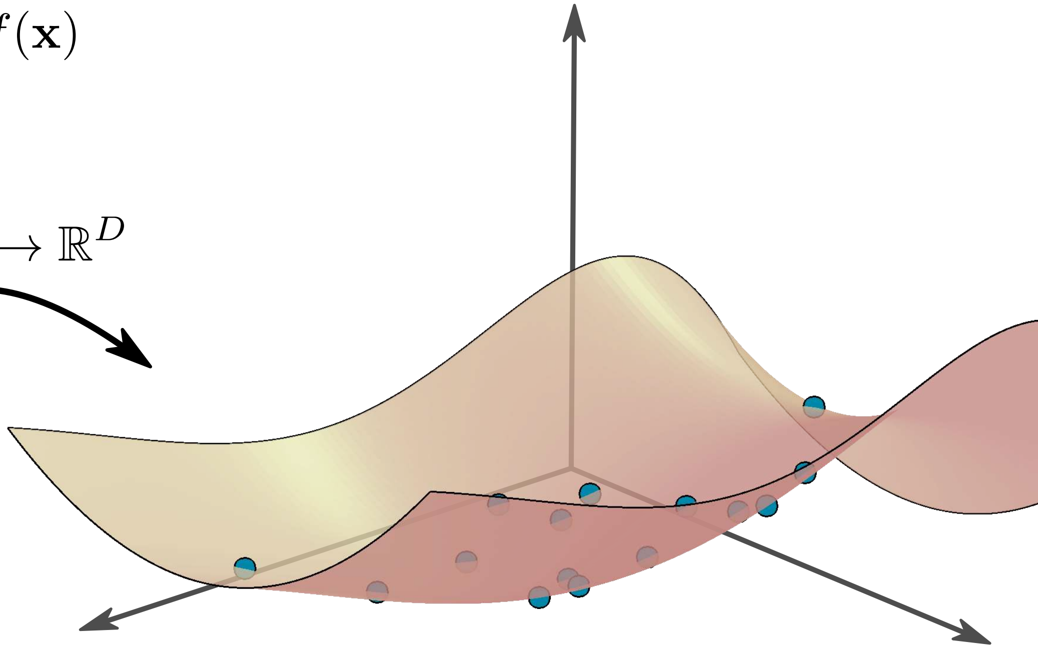


(a) Seed = 123 (b) Seed = 1234

(c) Seed = 12345 (d) Seed = 123456

$f : \mathbb{R}^d \to \mathbb{R}^D$



Too much flexibility in the model (e.g. when using neural nets) allow us to reparametrize the latent representation without changing model fit. Renders latent representations unstable and unsuitable for posthoc processing/interpretation.

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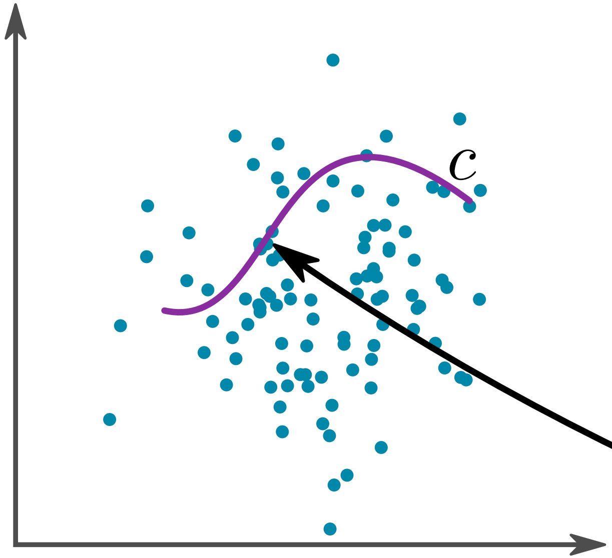

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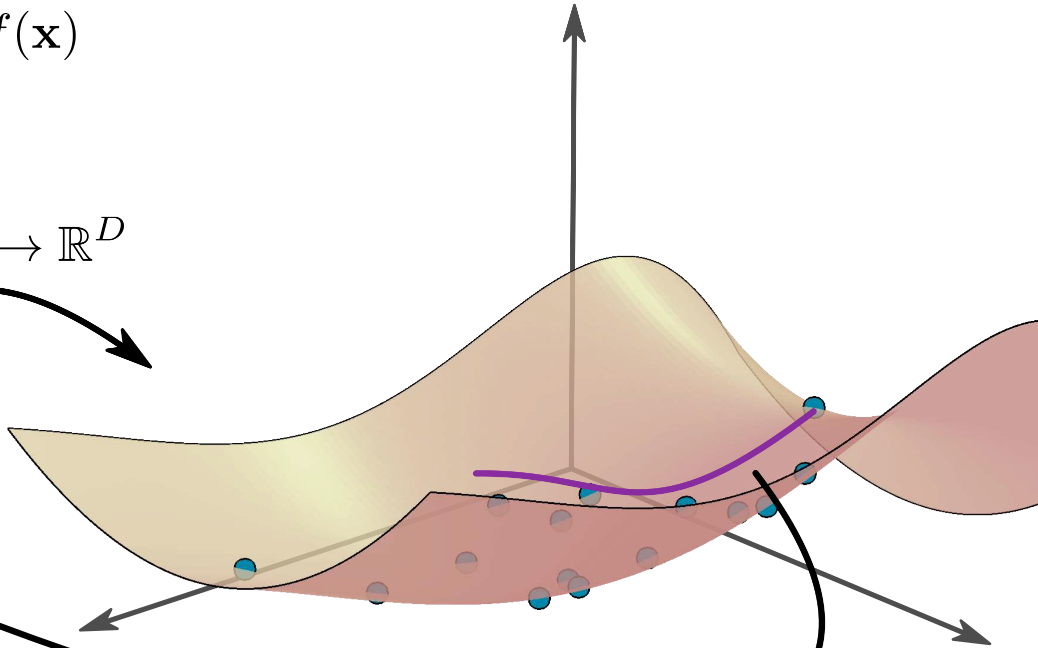

Define

$$\text{Length}(c) = \int_0^1 \left\| \frac{\mathrm{d}}{\mathrm{d}t} f(c_t) \right\| \mathrm{d}t$$

The shortest path and associated distance is then

$$c_{01} = \operatorname*{argmin}_c \text{Length}(f(c))$$

$$\text{dist}(c_0, c_1) = \text{Length}(c_{01})$$

Define length of *latent curve* to be the length of the *decoded curve*.
From that definition, you just go where the math takes you...

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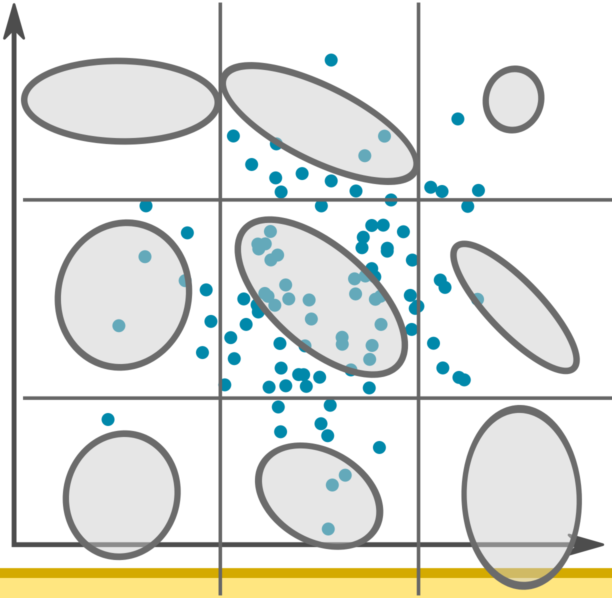

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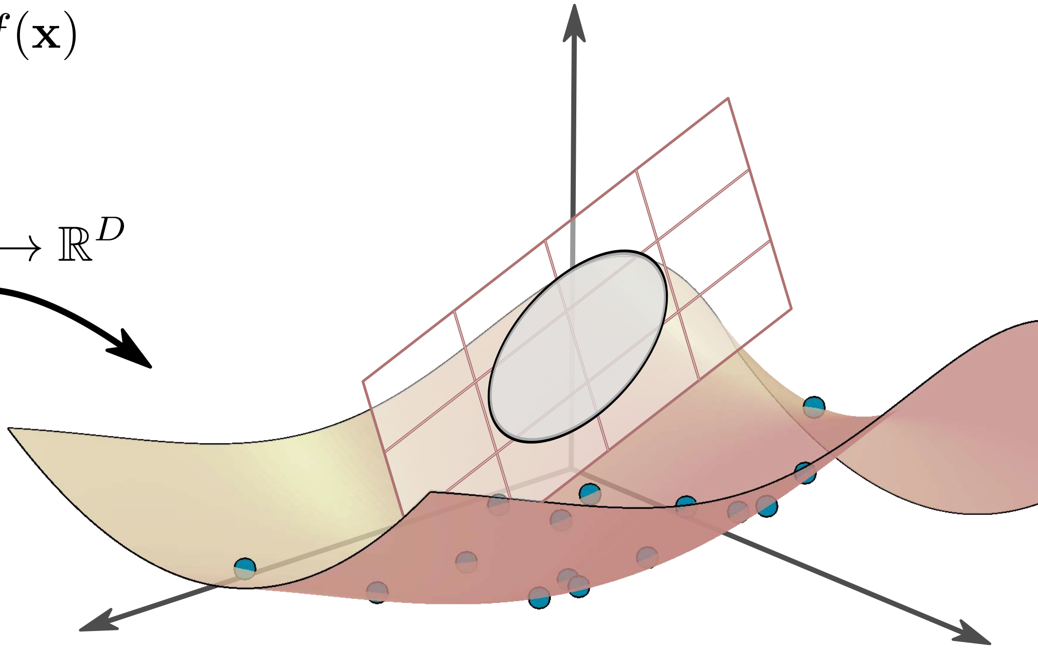

Locally, we can measure inner products

$$\langle \delta_1, \delta_2 \rangle_{\mathbf{x}} = (\mathbf{J}_{\mathbf{x}} \delta_1)^{\intercal} (\mathbf{J}_{\mathbf{x}} \delta_2) = \delta_1^{\intercal} \boxed{\mathbf{J}_{\mathbf{x}}^{\intercal} \mathbf{J}_{\mathbf{x}}} \delta_2$$

We call $\boxed{\mathbf{G}_{\mathbf{x}} = \mathbf{J}_{\mathbf{x}}^{\intercal} \mathbf{J}_{\mathbf{x}}}$ *the (Riemannian) metric*

Locally, we can measure inner products in the tangent space of our manifold. This gives rise to a local inner product in the latent space: a Riemannian metric.

Goals of this module (week 5-8)

Week 5: chapters 1-5

- (lack of) identifiability of representations
- what are manifolds?
- distances on manifolds
- Riemannian metrics

Week 6: chapters 6-9

- Riemannian metrics
- geodesics (shortest paths)
- computing geodesics and distances
- addition and subtraction on manifolds

Week 7: chapters 11, 12, 14

- noise manifolds
- information geometry
- parameter manifolds

Week 8

- build a complete stack for identifiable computations in deep LVMs.
- if you do the weekly exercises, you will have done the bulk of the project.

You are here

I'm always looking for feedback

I need two volunteer "feedbackers"

Pullback metrics

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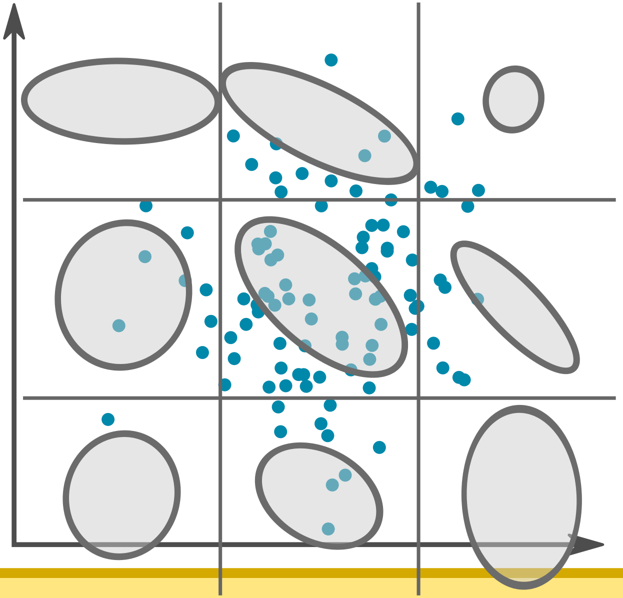

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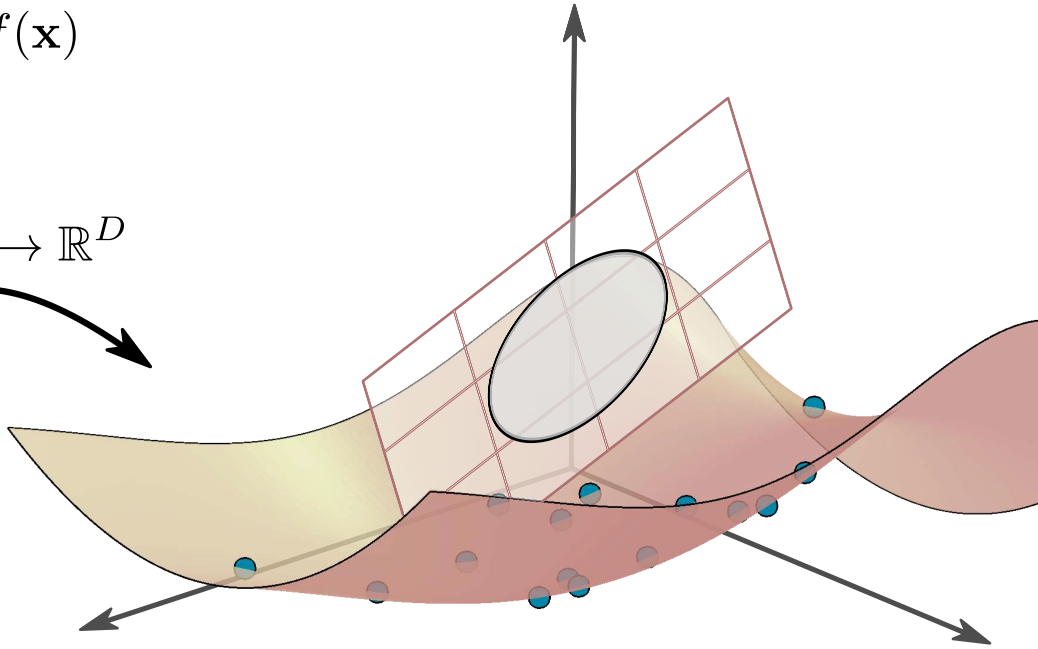

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This type of Riemannian metrics are known as "pullback metrics" and they are neat because they ensure reparametrization invariance.

But they are not the only option...

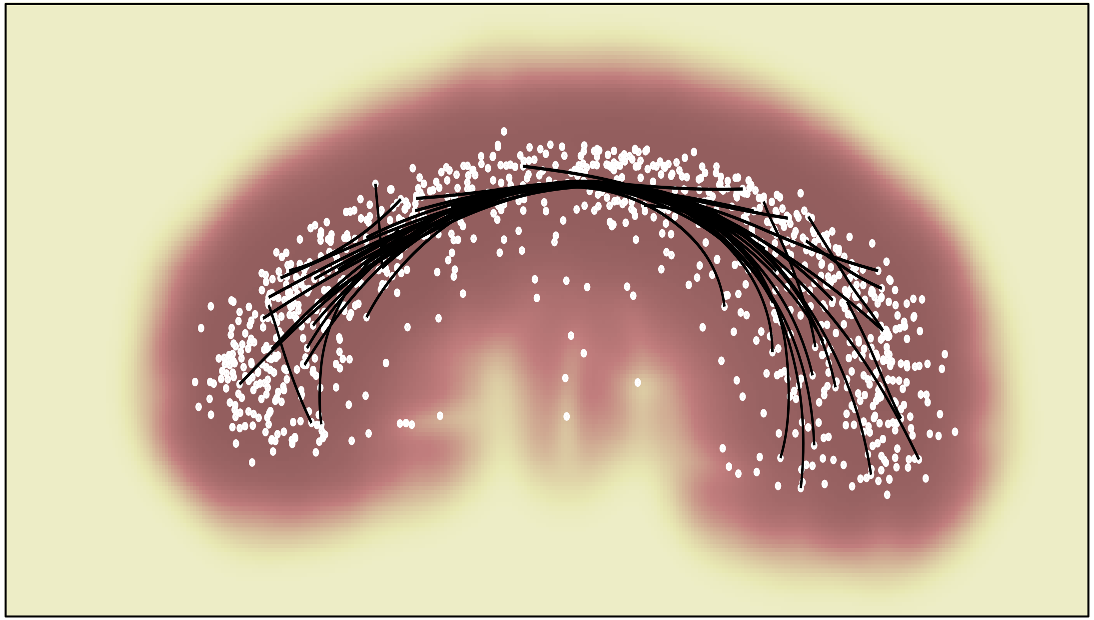
Abstract Riemannian metrics

Since calculations only requires access to the metric, we can also design these to satisfy whatever property we want.

For example, if we have access to a density, then we can choose a metric

$$\mathbf{G}_{\mathbf{y}} = g(\mathbf{y})\mathbf{I}, \qquad\qquad g(\mathbf{y}) = \frac{1}{p(\mathbf{y})}.$$

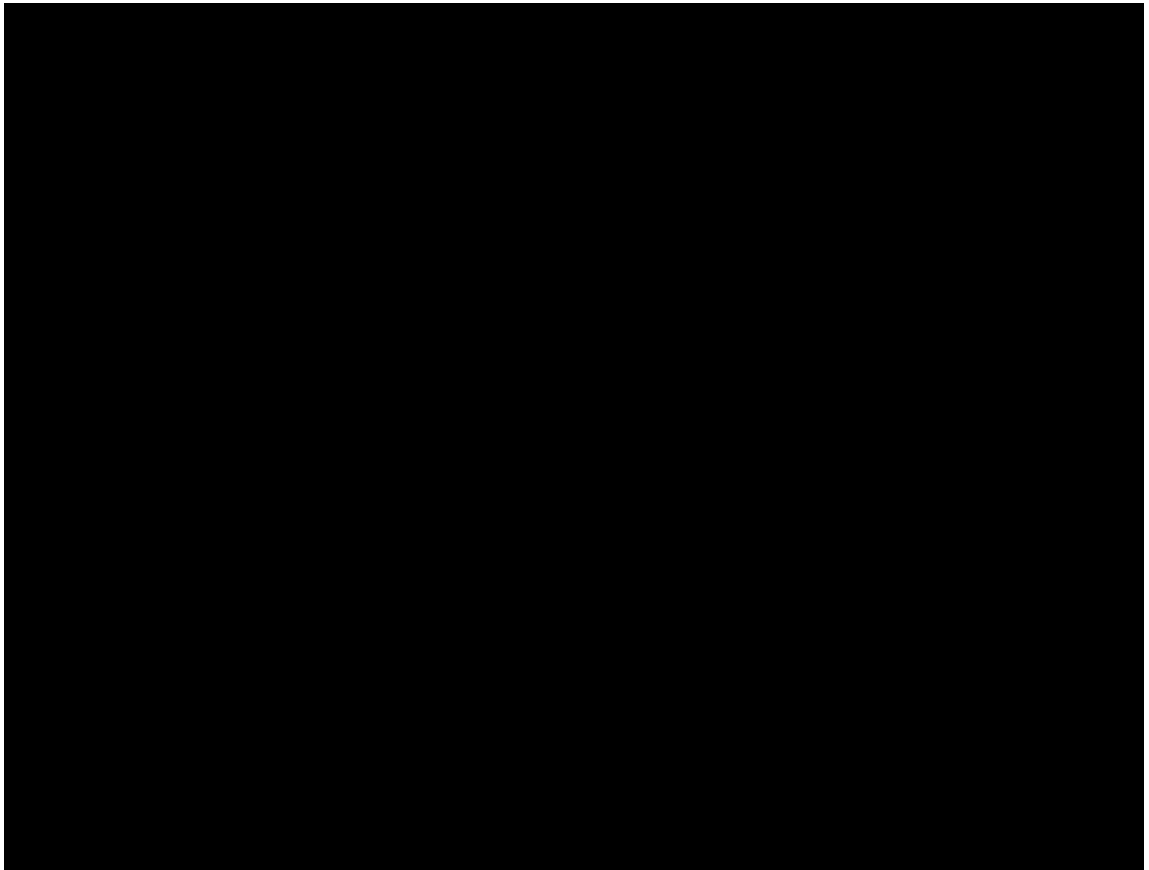
Now shortest paths roughly follow the trend of the data (useful e.g. for interpolation)



Abstract Riemannian metrics

We can make a robot avoid obstacles (the metric is applied to the robot hand position)

$$\mathbf{G}_{\mathbf{y}} = \left(1 + \eta \exp\left(\frac{-\|\,\mathbf{y} - \boldsymbol{o}\|^2}{2r^2}\right)\right)\mathbf{I}_3, \quad \mathbf{y} \in \mathbb{R}^3$$



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Geodesics

(aka shortest paths)

Shortest paths

We have seen that we can use shortest paths to define distances. Later we'll see how we can use such paths for generalizing addition and subtraction. But first, let's have a closer look at these shortest paths.

Recall that the shortest path between two points $c_0, c_1 \in \Omega$ is

$$c_{01} = \underset{c}{\operatorname{argmin}} \operatorname{Length}(f(c))$$

(technically it should be an infimum...)

$$\text{subject to } c(0) = c_0 \text{ and } c(1) = c_1$$

Geodesics

For historical reasons, we often refer to shortest paths as *geodesics*.
(*Geodesy* is the art of making measurements about planet Earth.
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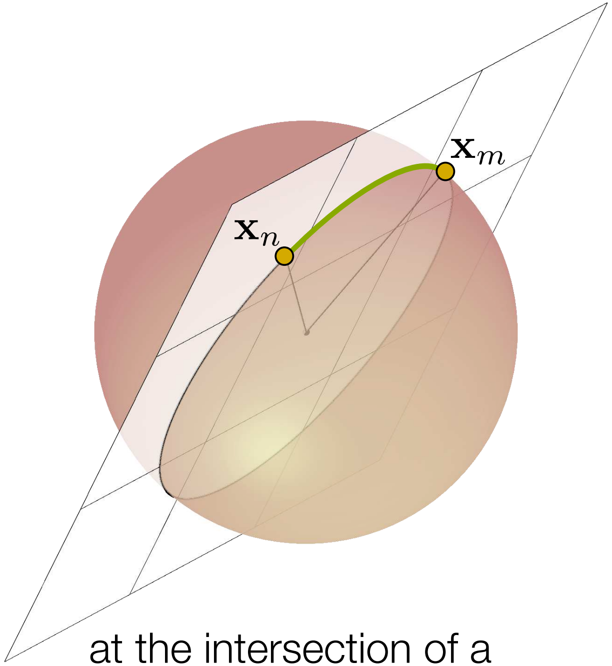

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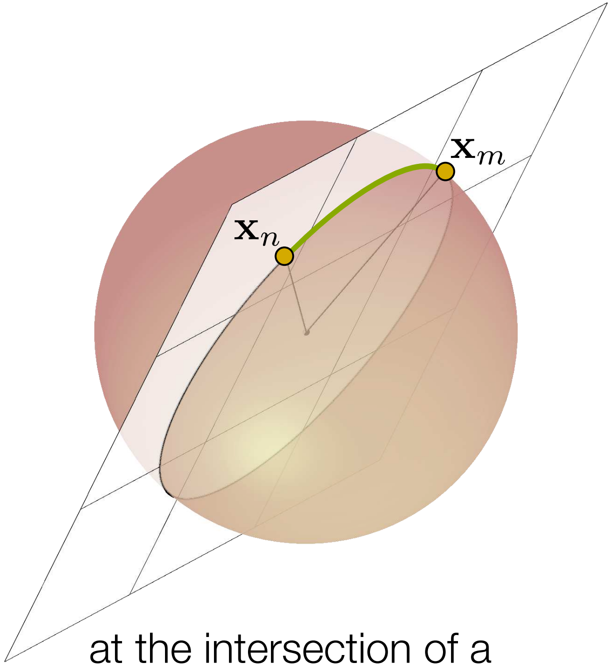

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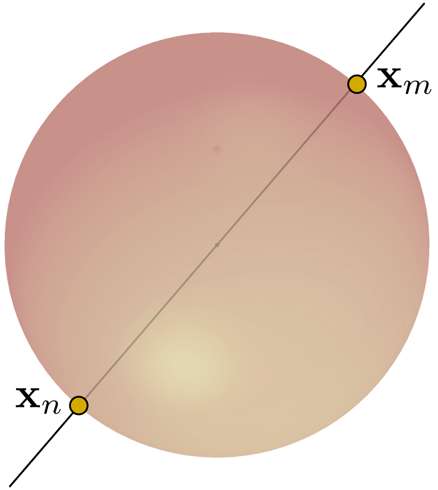

which is not defined for antipodal points;

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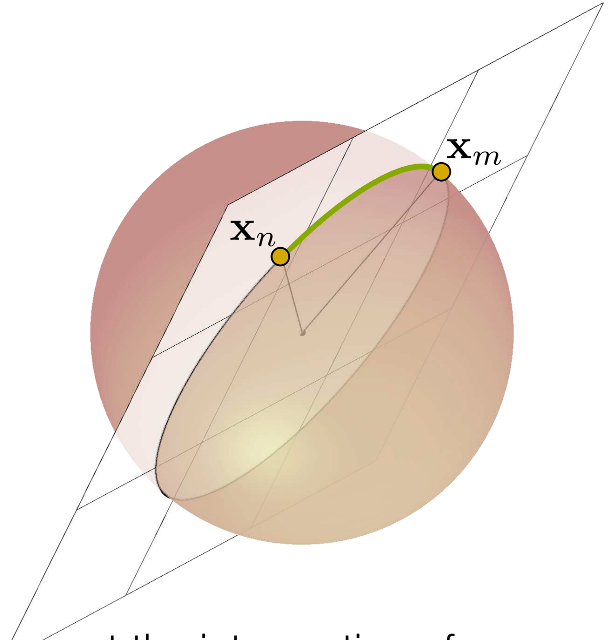

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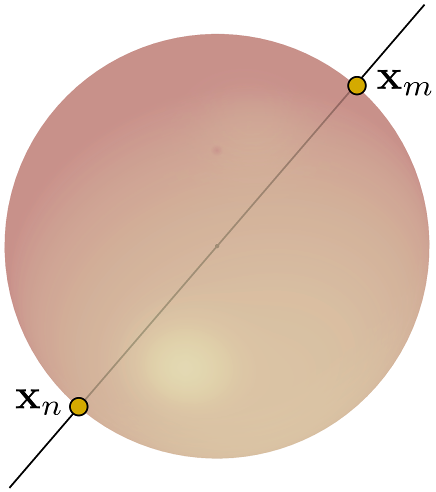

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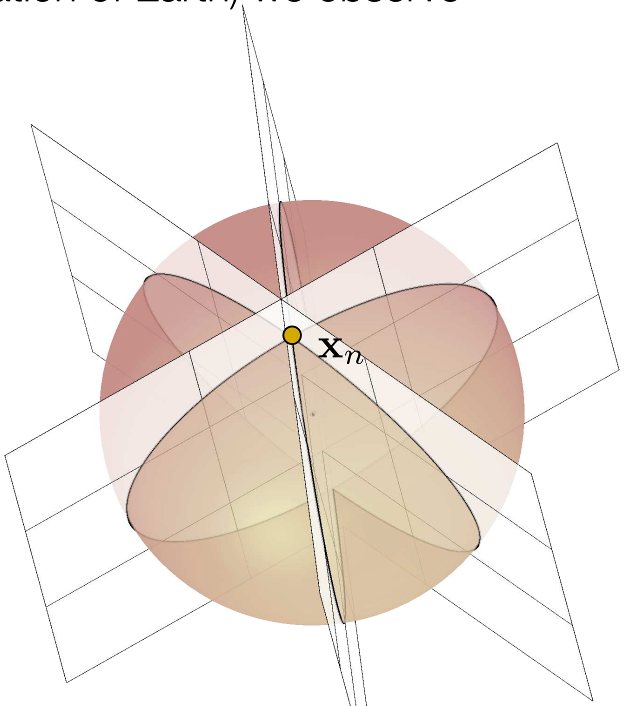

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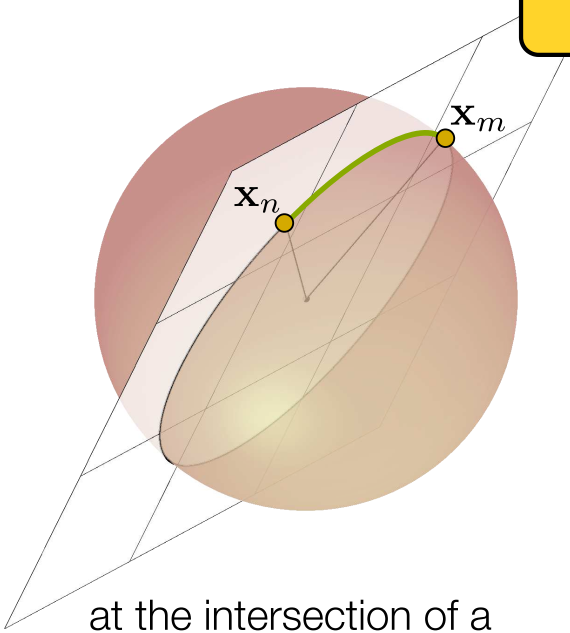
Take-home message:
Don't expect shortest paths to be unique!



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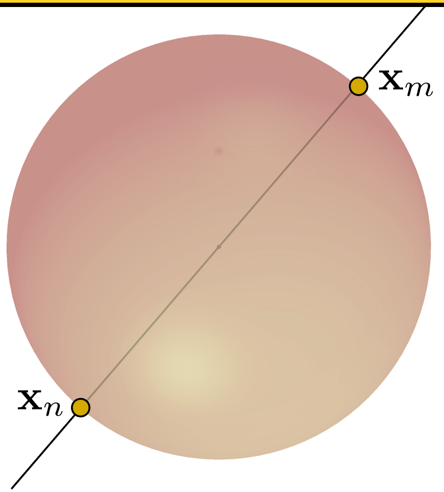

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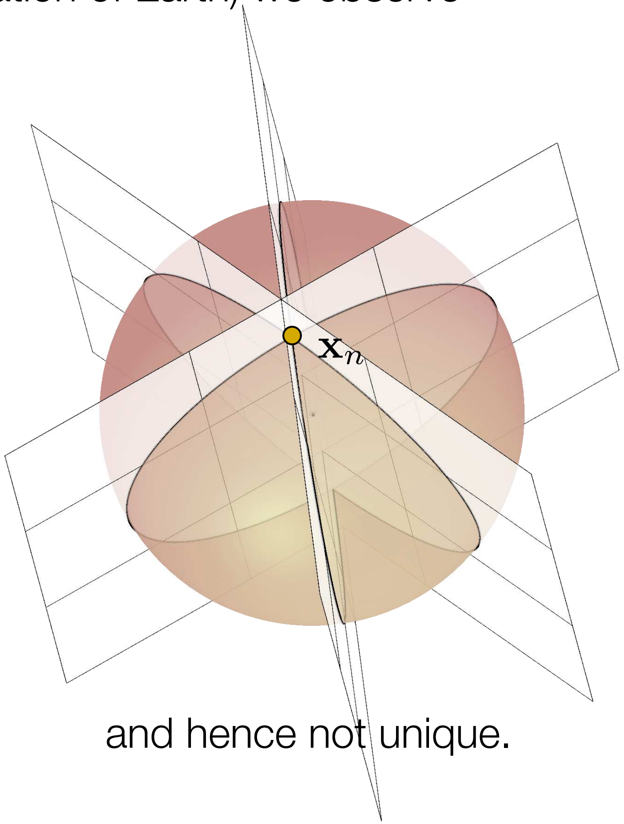

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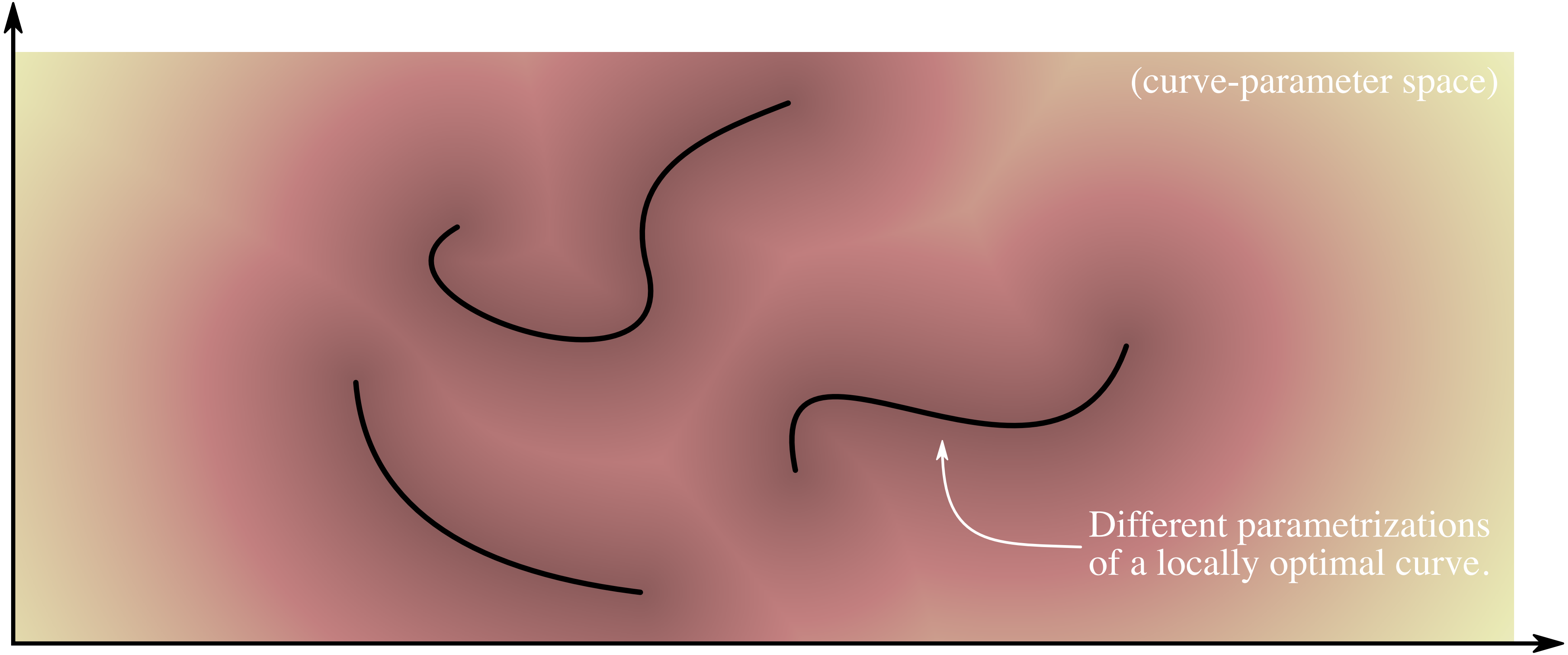
Geodesics: constant speed parametrizations

We have previously seen that we may reparametrize any given curve in a multitude of ways without changing its shape. In practice, we'll have to *optimize* in order to recover a geodesic. Reparametrization then implies that shortest paths are not even *locally unique* in whichever parameters we use to represent curves. Instead, shortest paths form continuums in this parameter space:



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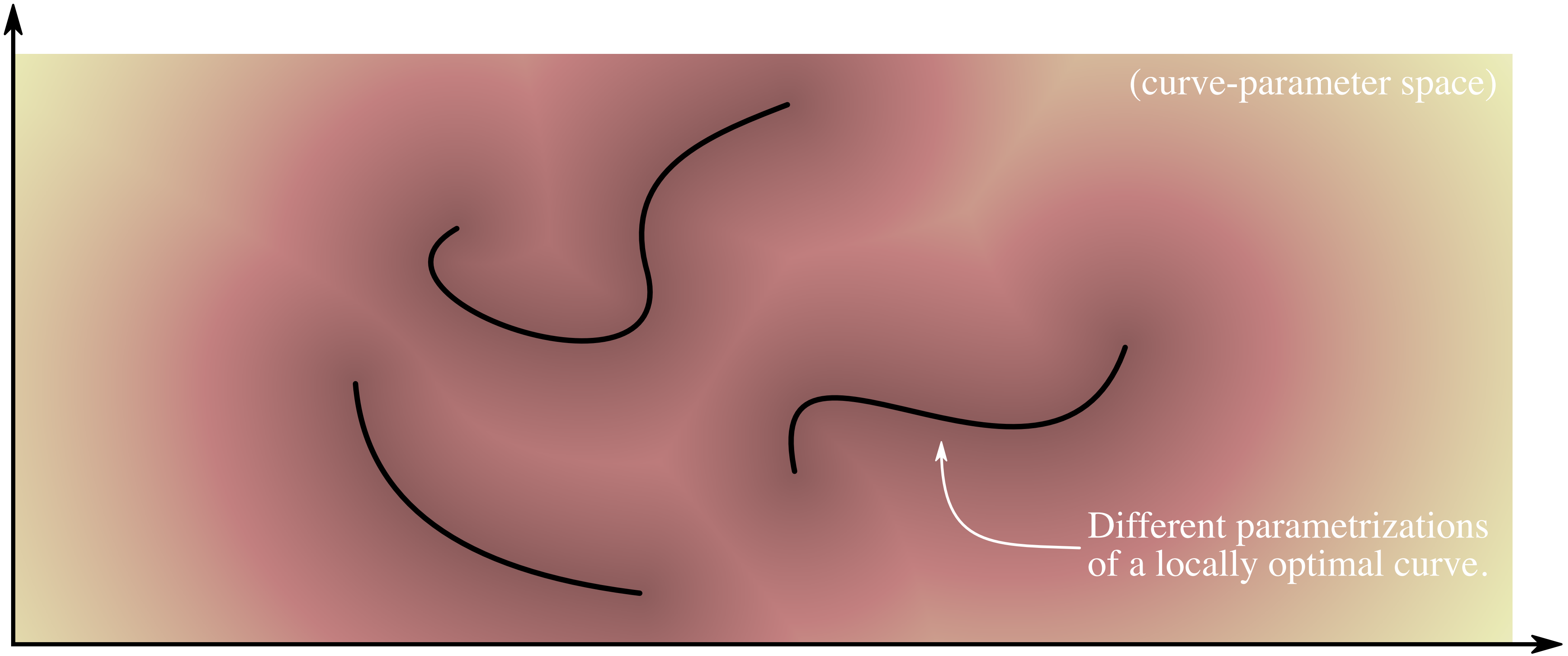


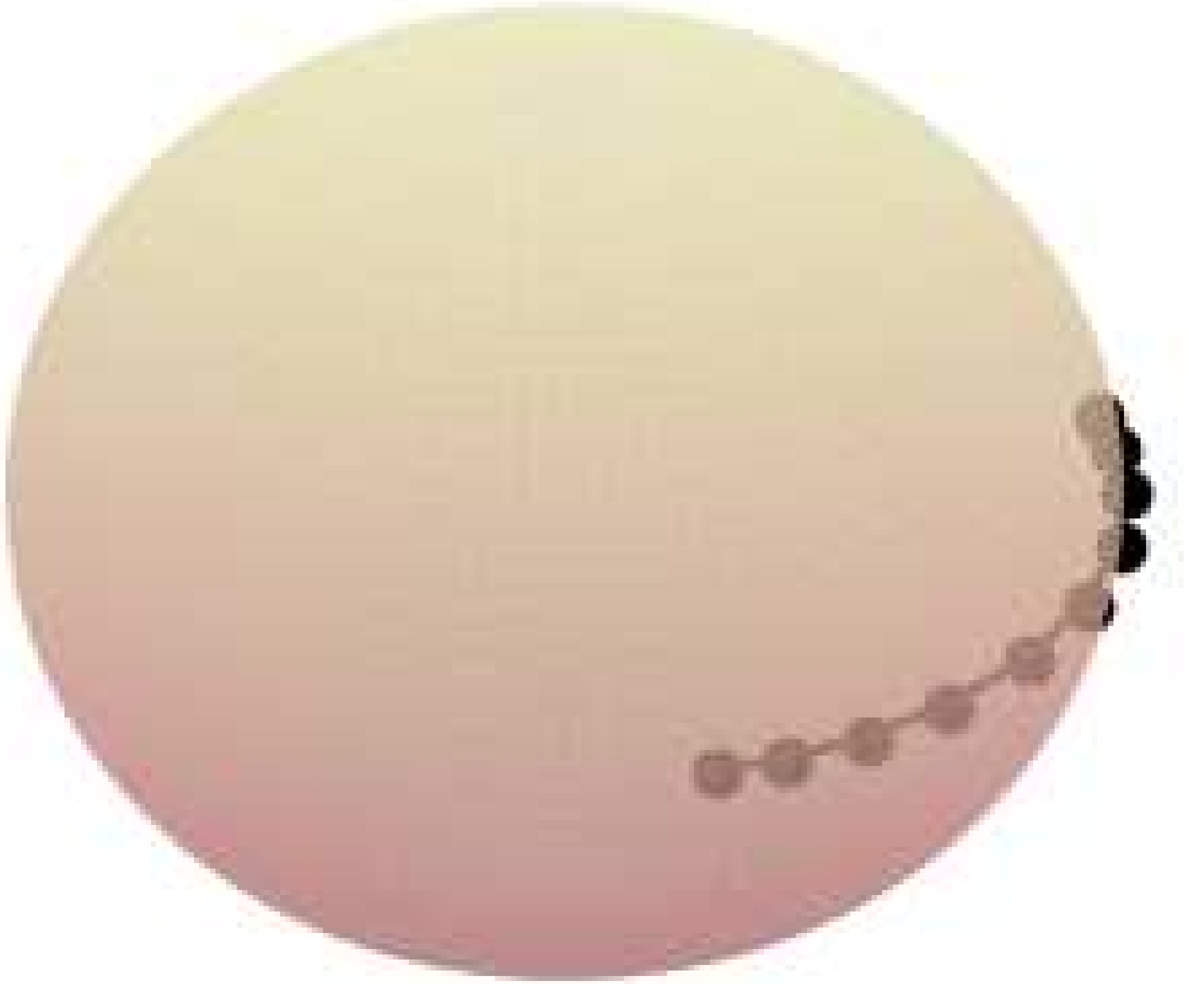
One way to make the parametrization unique is to choose a parametrization. There really is no canonical choice of curve parametrization, but an often-convenient choice is the *constant-speed parametrization*. That is, a parametrization chosen such that $||\dot{c}_t||$ is constant (independent of t). Numerically, this amounts to equidistant points along the curve.

What happens in practice?

I have discretized a curve that connects two points. I have chosen this curve to be a shortest path between the points.

Now I minimize the length of this curve while keeping end-points fixed and ensuring that the curve stays on the manifold.

What happens? Discuss...



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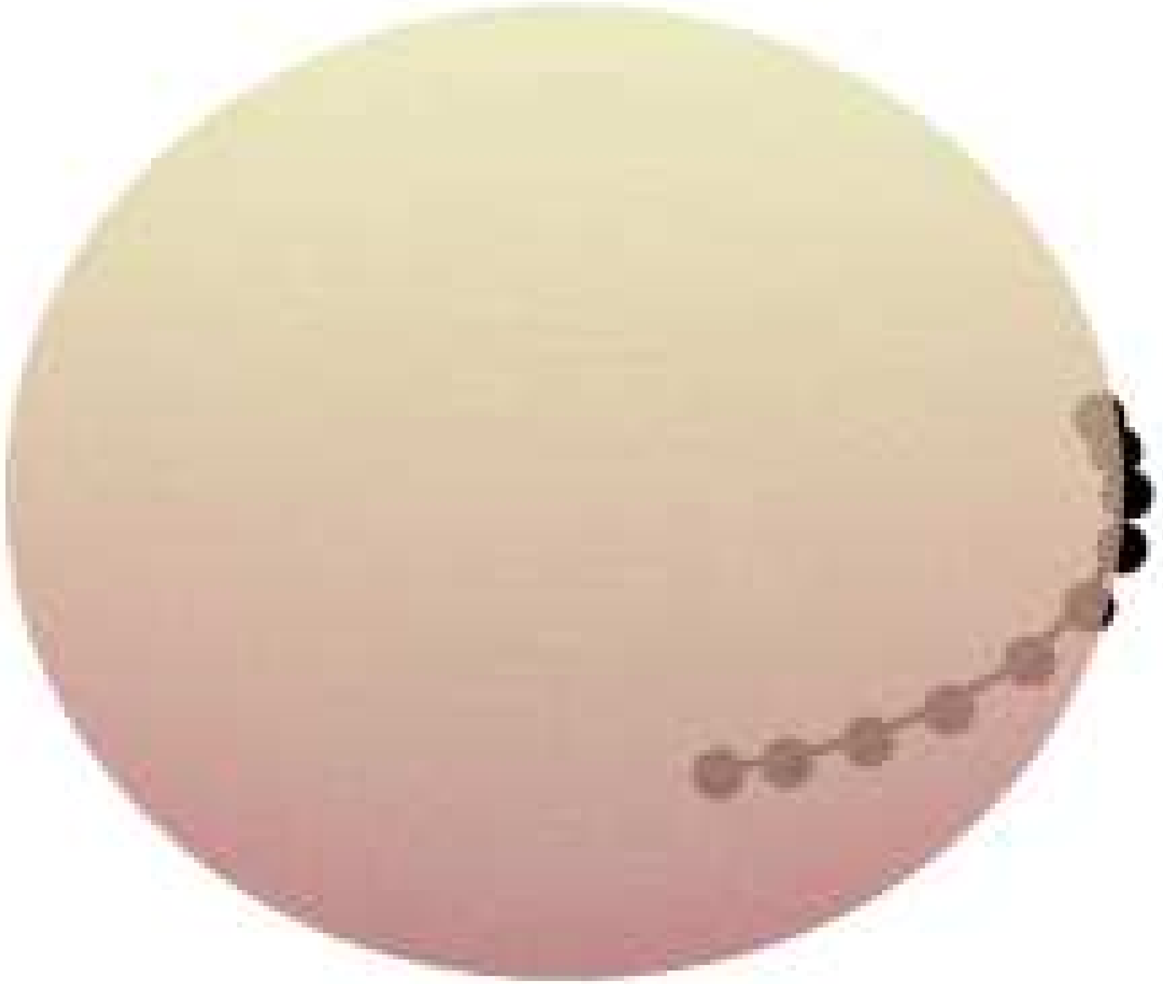

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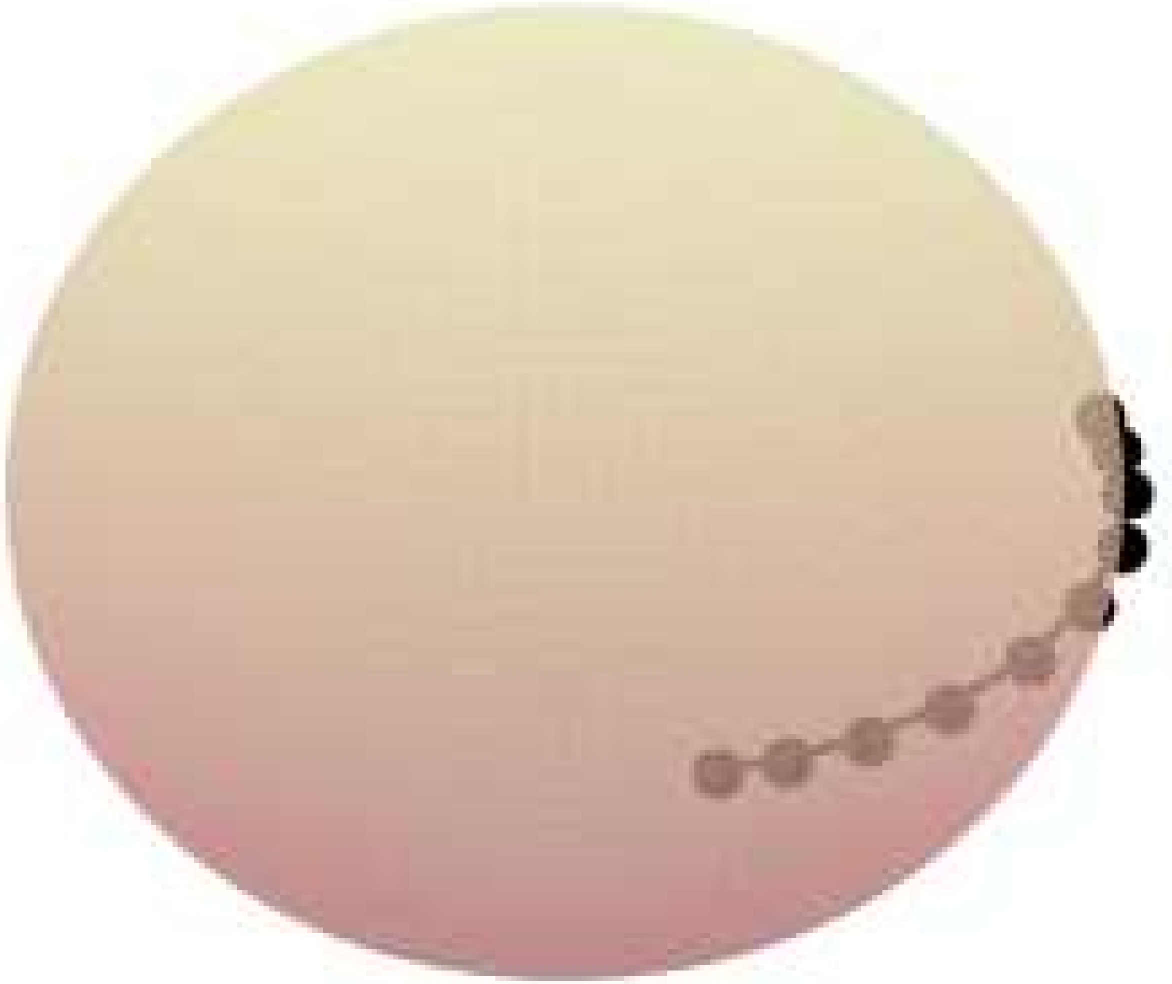

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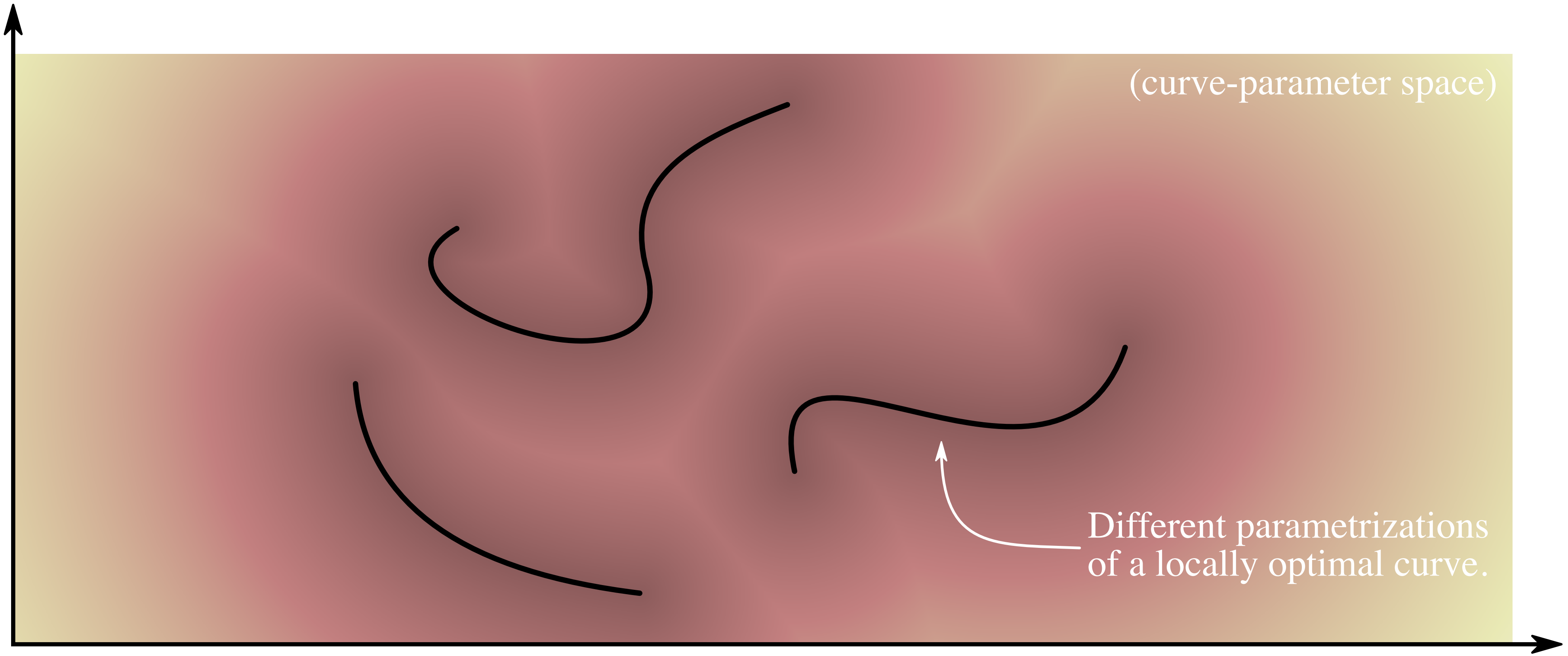

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If we let $v = \|\dot{c}_t\|$ denote this constant, then the length of a curve $c : [0, 1] \rightarrow \mathbb{R}^D$ simplify to

$$\begin{aligned}\text{Length}[c] &= \int_0^1 v \mathrm{d}t \\ &= v \int_0^1 \mathrm{d}t = v.\end{aligned}$$

That is, to evaluate the length of a constant speed curve, we only need to evaluate its velocity once rather than perform integration. That's rather neat!

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But how do we get constant speed geodesics?

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Curve energy

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Recall the Cauchy-Schwartz inequality

$$|\langle u, v\rangle| \leq \|u\| \; \|v\|,$$

where equality holds if and only if u and v are parallel.
This holds in any inner product space assuming $\|u\| = \sqrt{\langle u, u\rangle}$.

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We will look at curves, and hence need an inner product between curves.
We'll make the usual choice

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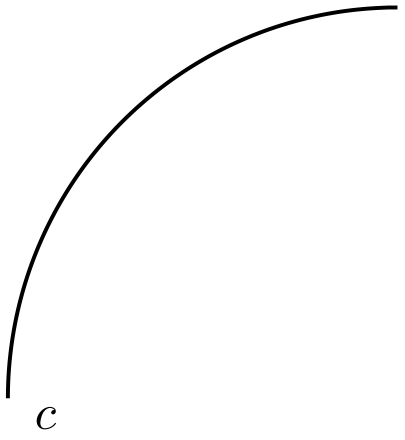

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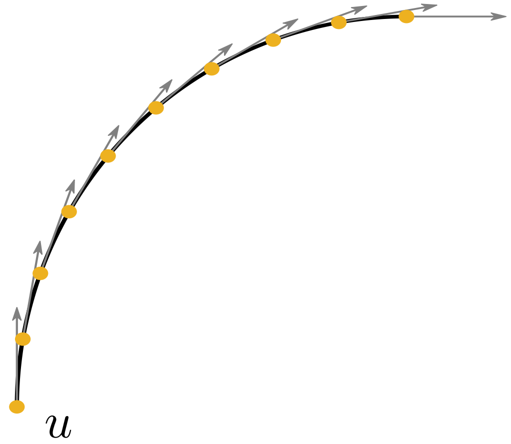

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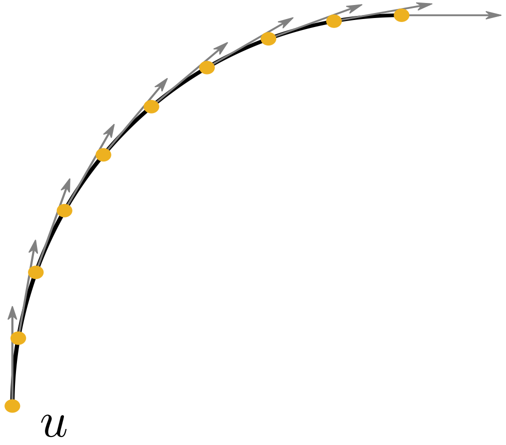

To apply Cauchy-Schwarts we introduce a seemingly silly curve

$$v : [0, 1] \to \mathbb{R}$$
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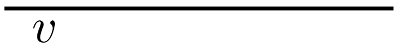

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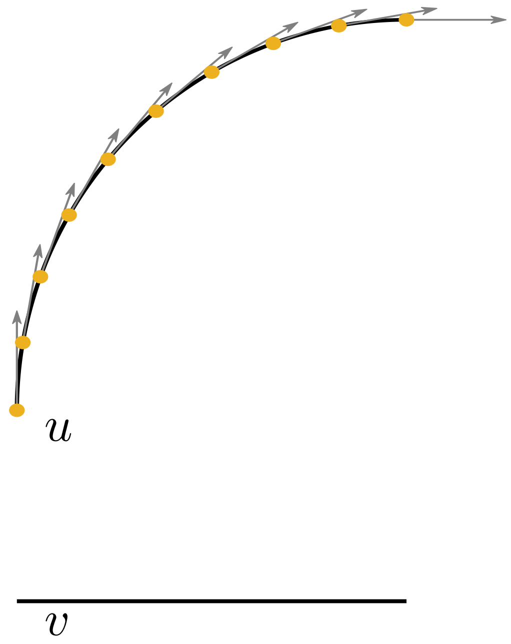

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We can then write the length of c in terms of u and v

$$\mathrm{Length}[c] = \int_0^1 \|\dot{c}_t\| \mathrm{d}t = \langle u, v \rangle.$$

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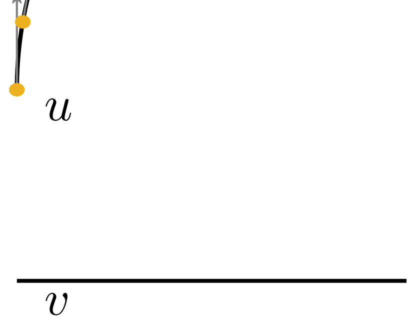

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Applying Cauchy-Schwarts $|\langle u, v \rangle| \leq \|u\| \; \|v\|$ then gives

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$$u : [0,1] \to \mathbb{R}$$
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To apply Cauchy-Schwarts we introduce a seemingly silly curve

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The inequality is a direct application of the Cauchy-Schwartz in equality.
We know that equality is satisfied if and only if u and v are parallel, i.e.

$$u_t = \alpha \ v_t \quad \Leftrightarrow \quad \|\dot{c}_t\| = \alpha.$$

In other words, the inequality becomes an equality if and only if $\|\dot{c}_t\|$ is constant
(c has a constant speed parametrization)

Curve energy

Here's what we know:

In general, we have the bound

$$\mathrm{Length}^2[c] \leq \mathcal{E}[c]$$

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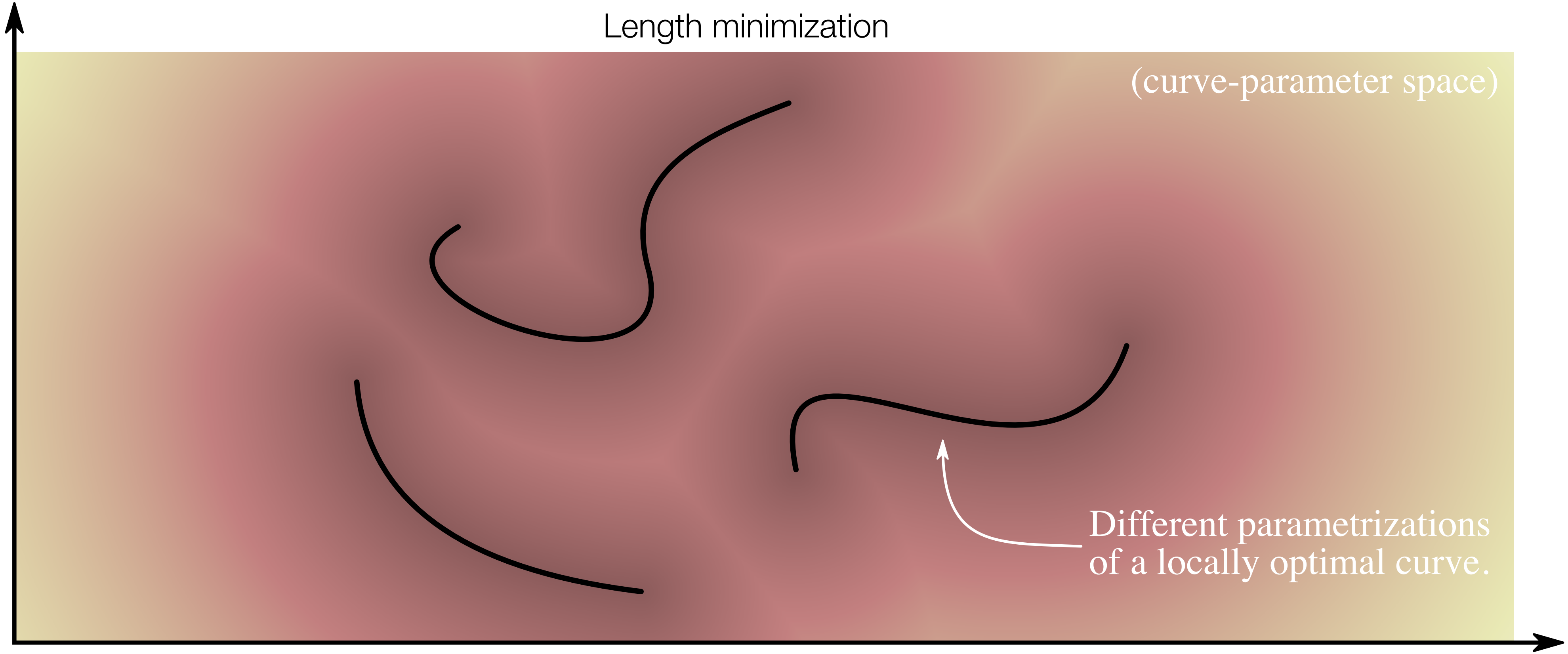

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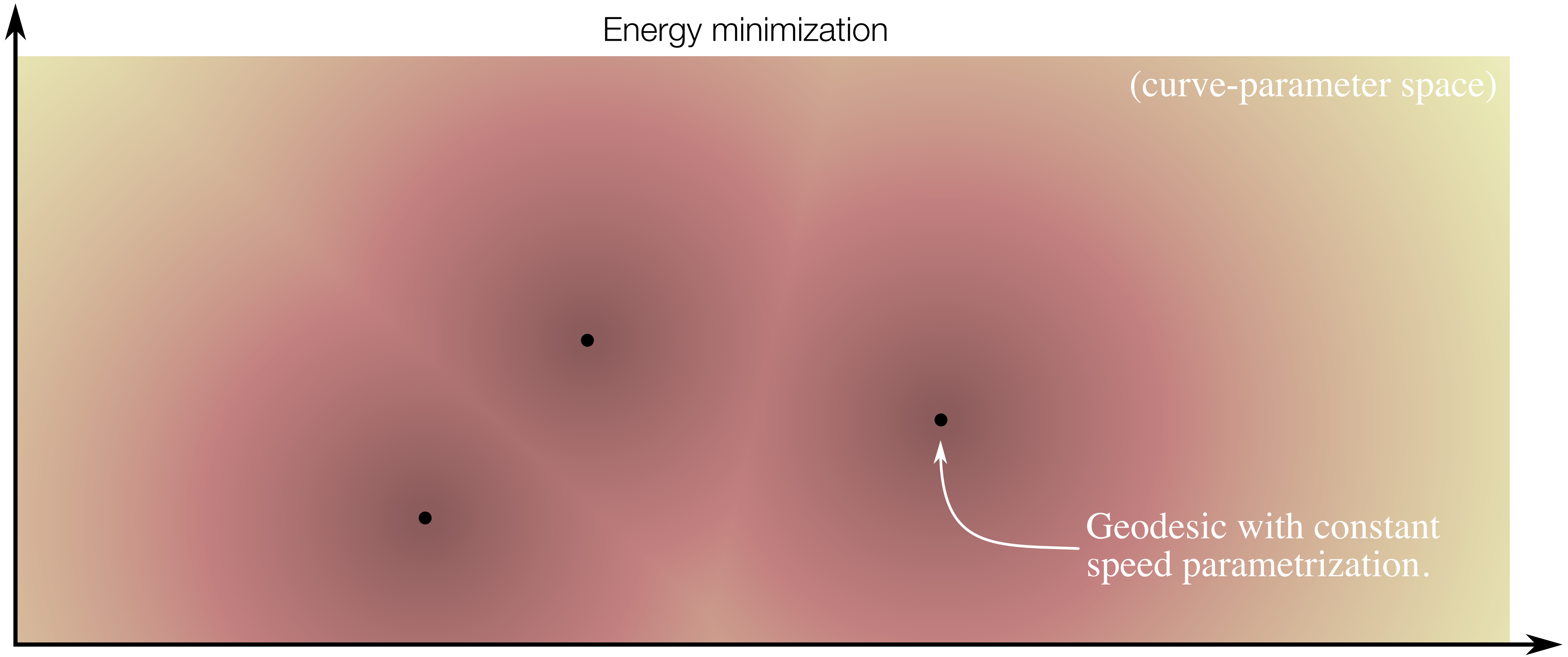


Computing Geodesics

Conceptually easy, practically.... well...

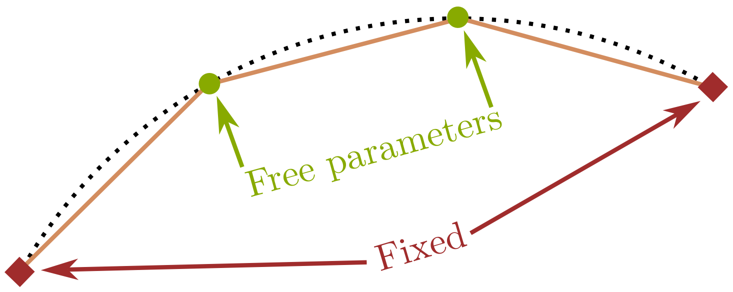
We have a simple recipe for computing geodesics numerically:

Compute connecting geodesic

1. parametrize a curve that connects c_0 and c_1,
2. numerically approximate the energy of the parametric curve, and finally
3. minimize the energy using gradient-based optimization.

Choice #1: how to parametrice the curve

The simplest option is a piecewise linear curve.
It's easy to keep the end-points fixed (just don't optimize them), and it's easy to figure out what's going on.



Alternatives includes polynomial curves, splines, neural networks, etc.

Choice #2: how to approximate the curve energy

This boils down evaluating an integral numerically somehow,

$$\mathcal{E}[c] = \int_0^1 \|\dot{c}_t\|^2 \mathrm{d}t.$$

If the metric is induced by a decoder, the simplest is to note that

$$\mathcal{E} = \int_0^1 \left\| \frac{\mathsf{d}}{\mathsf{d}t} f(c_t) \right\|^2 \mathsf{d}t. \qquad \tilde{\mathcal{E}}[c] = \sum_{s=1}^{S} \|f(c_s) - f(c_{s-1})\|^2.$$

We can't do this if we only have access to the metric (and not the decoder).
A benefit of using the decoder is that it's less taxing on the autodiff engine.

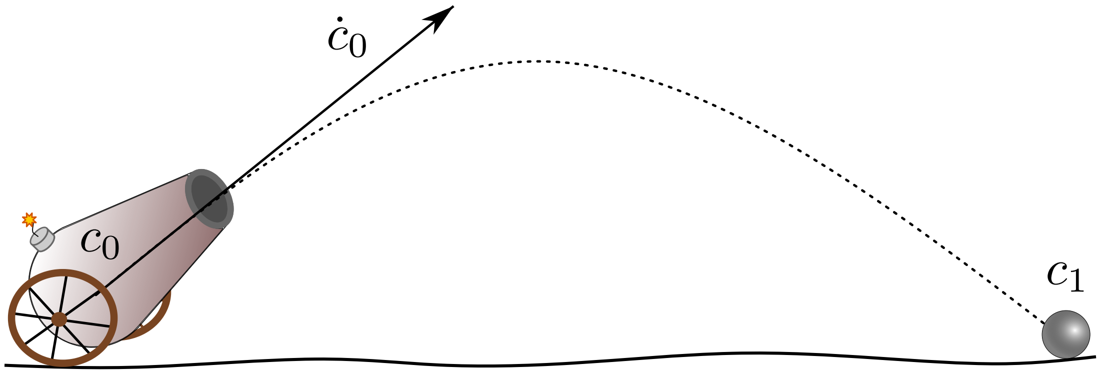
(see alternative strategies in the book)

Choice #3: how to optimize the energy

Practically speaking, most gradient-based optimizers works just fine.
In pytorch, I prefer SGD, Adam, or LBFGS. The latter is almost always the best, but it can be slower for easy problems.

You'll be playing with all of this stuff later....

Shooting Geodesics



Geodesic ODEs

To better understand geodesics we will look at a set of ordinary differential equations (ODEs) that govern their behavior.

First, we recall that geodesics are energy-minimizing curves, i.e. they minimize

$$\int_0^1 E(t, c_t, \dot{c}_t)\mathrm{d}t \qquad\qquad E : \mathbb{R} \times \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}$$

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$$\Gamma_j^{ik}(c_t) = \frac{1}{2} \left[\mathbf{G}_{c_t}^{-1}\right]_{ij} \left(2 \frac{\partial \left[G_{c_t}\right]_{ij}}{\partial \left[c_t\right]_k} - \frac{\partial \left[G_{c_t}\right]_{jk}}{\partial \left[c_t\right]_i} \right)$$

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Why is this even the least bit interesting?

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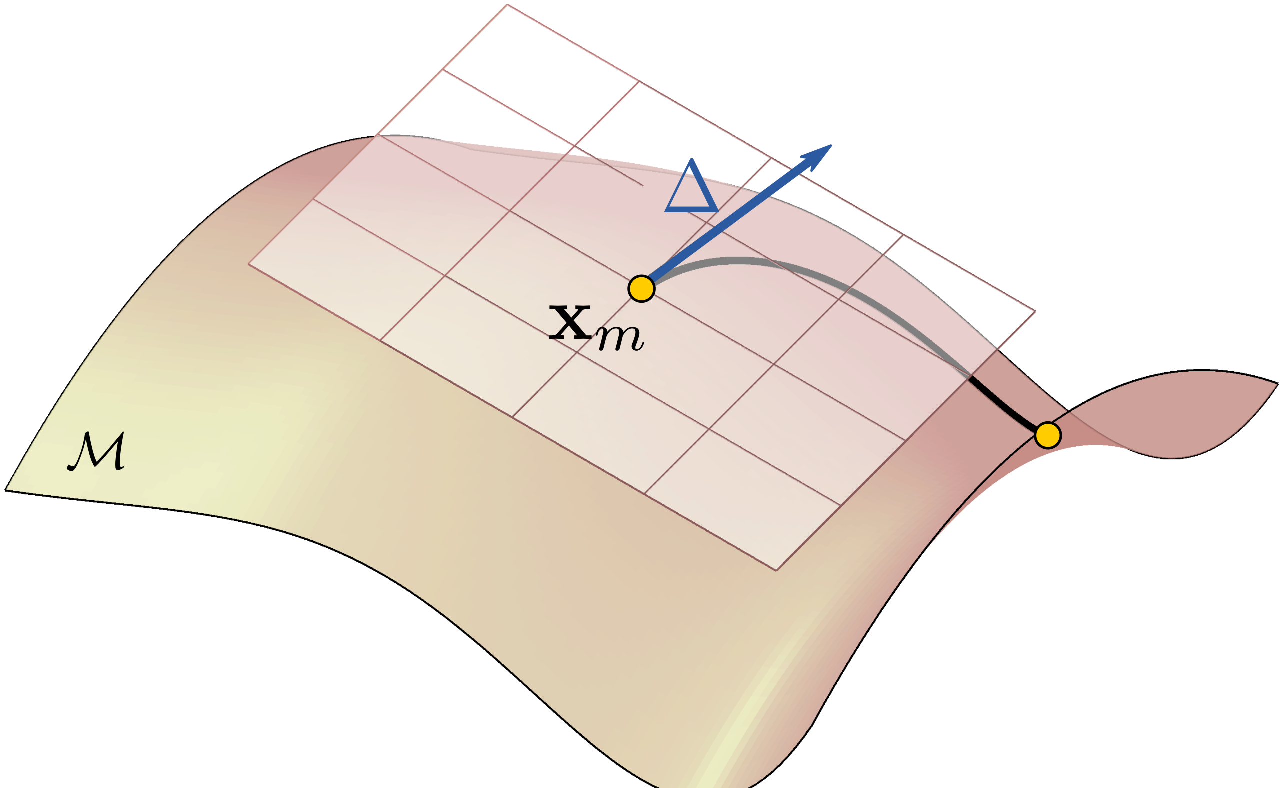
A new type of geodesic

So far we have considered geodesics that are defined by their end-points, i.e. paths when known starting and end-points.

But the ODE we can also solve by specifying an initial point and and an initial velocity. This is known as an *initial value problem* and the Picard-Lindelöf theorem tells us the resulting geodesic is actually unique (unlike when we specify start/end points).



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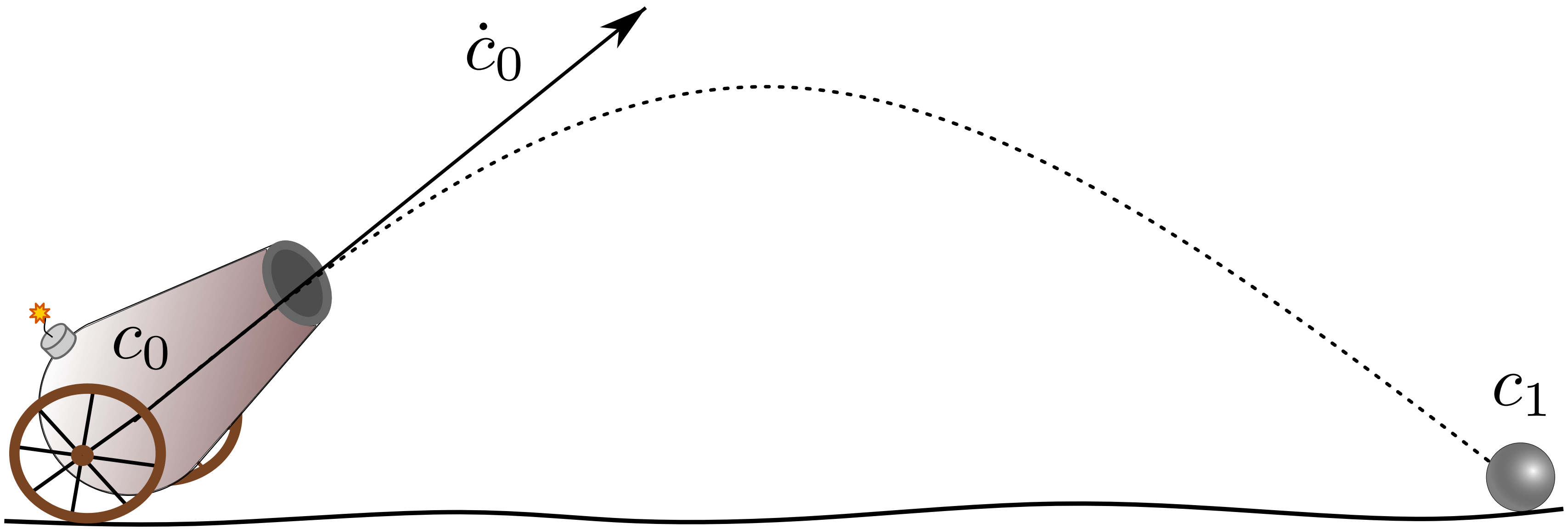
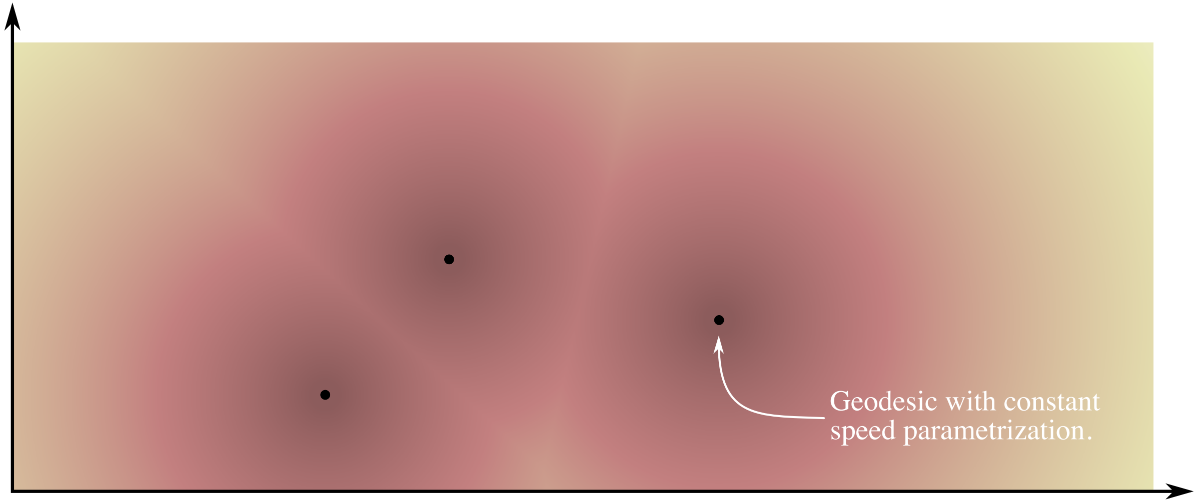


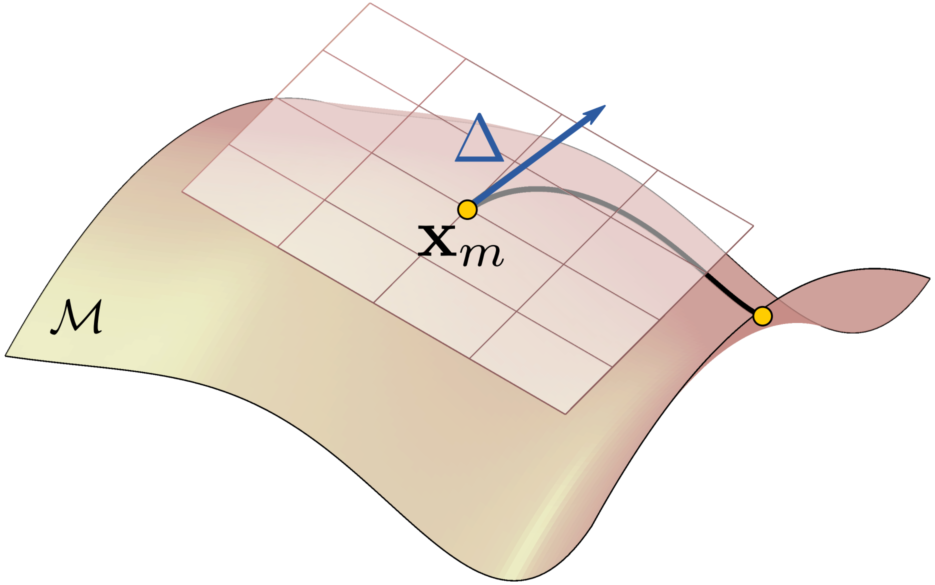
Intermediate summary

We can compute geodesics by minimizing curve energy

$$\mathcal{E}[c] = \int_0^1 \|\dot{c}_t\|^2 \mathrm{d}t.$$

This ensures well-defined optima corresponding to constant-speed curves.





Constant-speed geodesics follow a system of 2nd order ODEs

$$[\ddot{c}_t]_j = -\sum_{i=1}^{d}\sum_{k=1}^{d} \Gamma_j^{ik}(c_t)\,[\dot{c}_t]_k\,[\dot{c}_t]_j$$

This allow us to define geodesics from an initial point and an initial velocity vector.

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What exactly can we do with our `new data'?

We clearly cannot do Euclidean arithmetic (addition and subtract)
as this is not invariant to reparametrizations (you know, the identifiability issue).
But wouldn't it be nice to have something similar?

Euclidean arithmetic

Let us dig a bit into Euclidean arithmetic to better understand what it is we are asking for...

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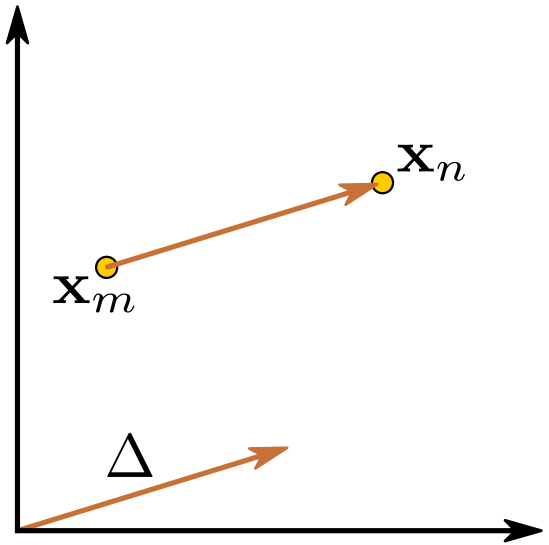

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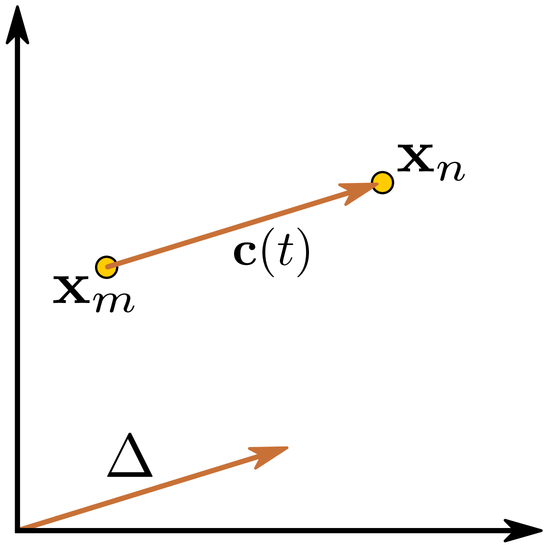
$$\mathbf{c}(t) = \Delta t + \mathbf{x}_m, \qquad t \in [0,1].$$



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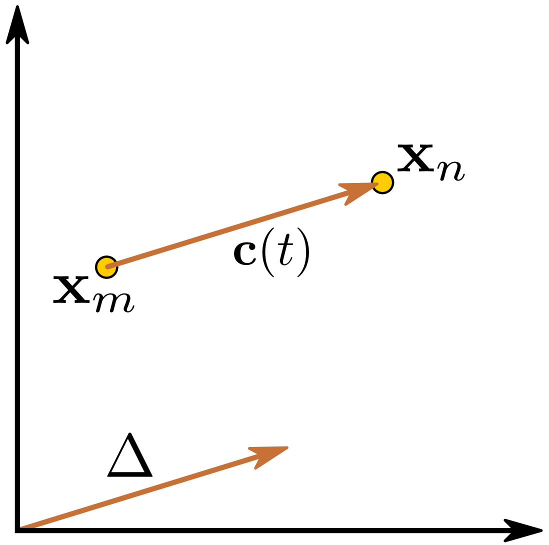
We see that subtraction is an operator which return a vector parametrizing the shortest path from one point to another.



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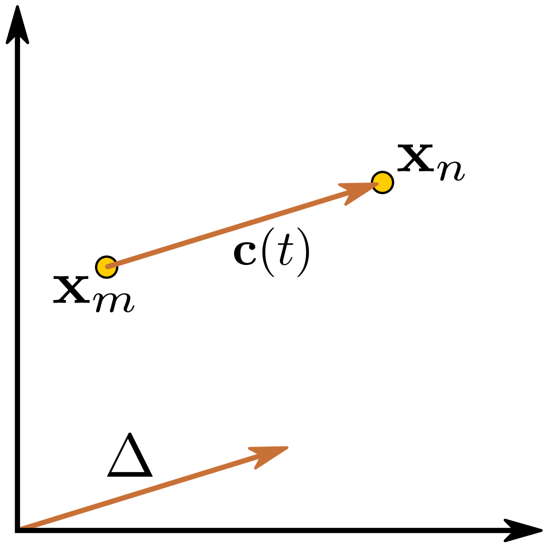
+ addition

Addition (plus) take a point $\mathbf{x}_m$ and a displacement vector Δ and return the point at the end of the shortest path corresponding to the displacement

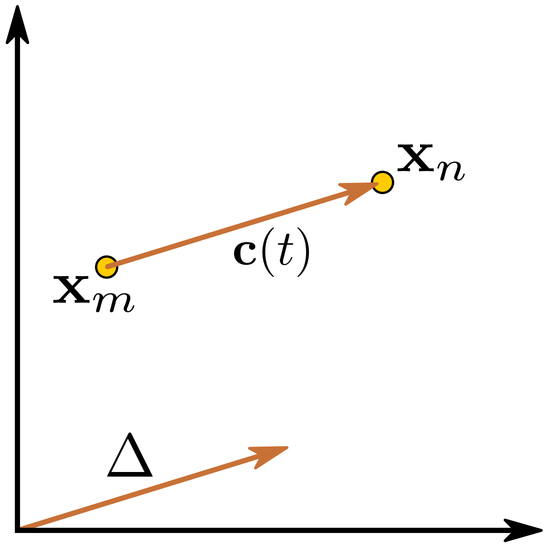
$$\mathbf{x}_n = \Delta + \mathbf{x}_m$$

(it's the inverse of subtraction)



Pointless abstractions?

Let's make matters painfully clear:
Thus far we have made a subtle distinction between *points* and *vectors*.
Let $\mathcal{M}$ denote the set of al points and let T denote the set of all vectors;
(note that in Euclidean geometry $\mathcal{M} = T = \mathbb{R}^D$)



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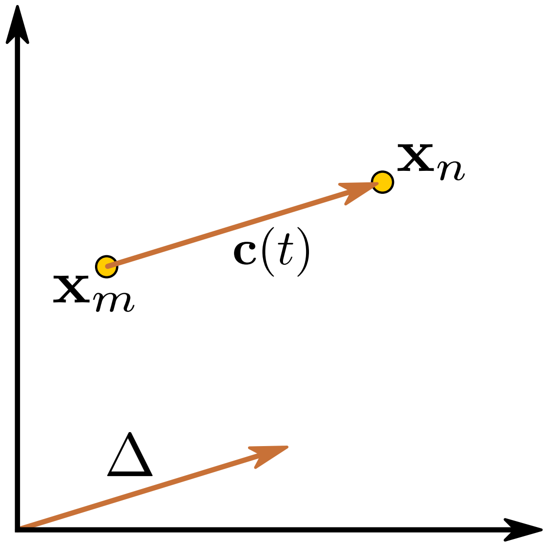

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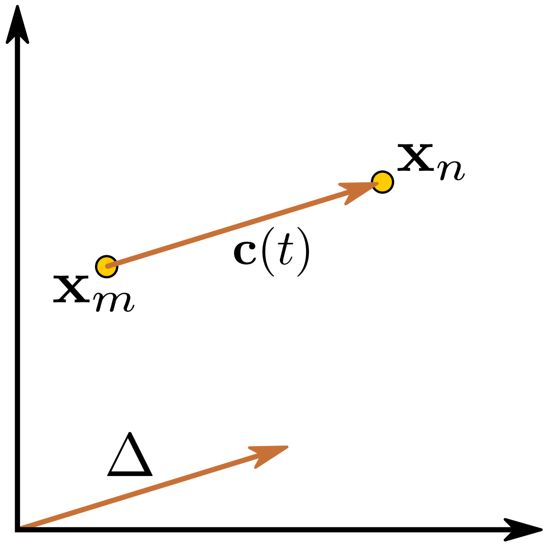
Finally, we note that the vector

$$\Delta = \mathbf{x}_n - \mathbf{x}_m$$

tell us how to get from one point to another, but also how far it is, i.e.

$$\mathrm{dist}(\mathbf{x}_n, \mathbf{x}_m) = \|\mathbf{x}_n - \mathbf{x}_m\| = \|\Delta\|.$$



Generalize!

Having settled when is even meant by addition and subtraction, we are now ready to generalize these operators to our learned Riemannian representations.

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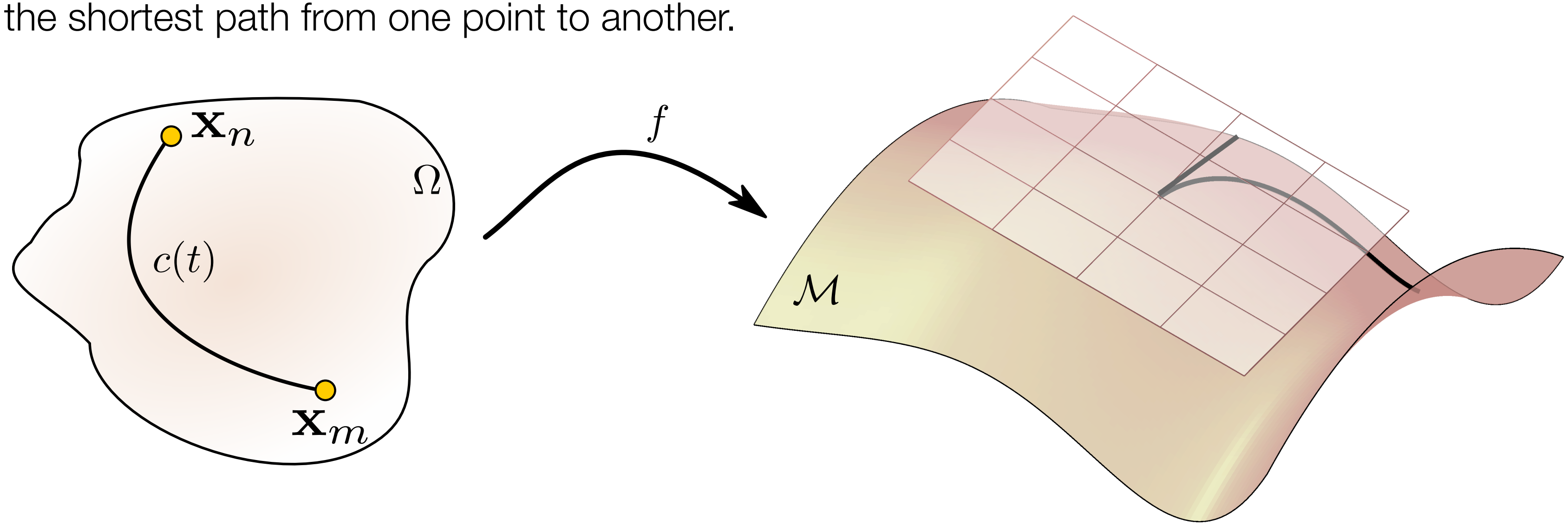

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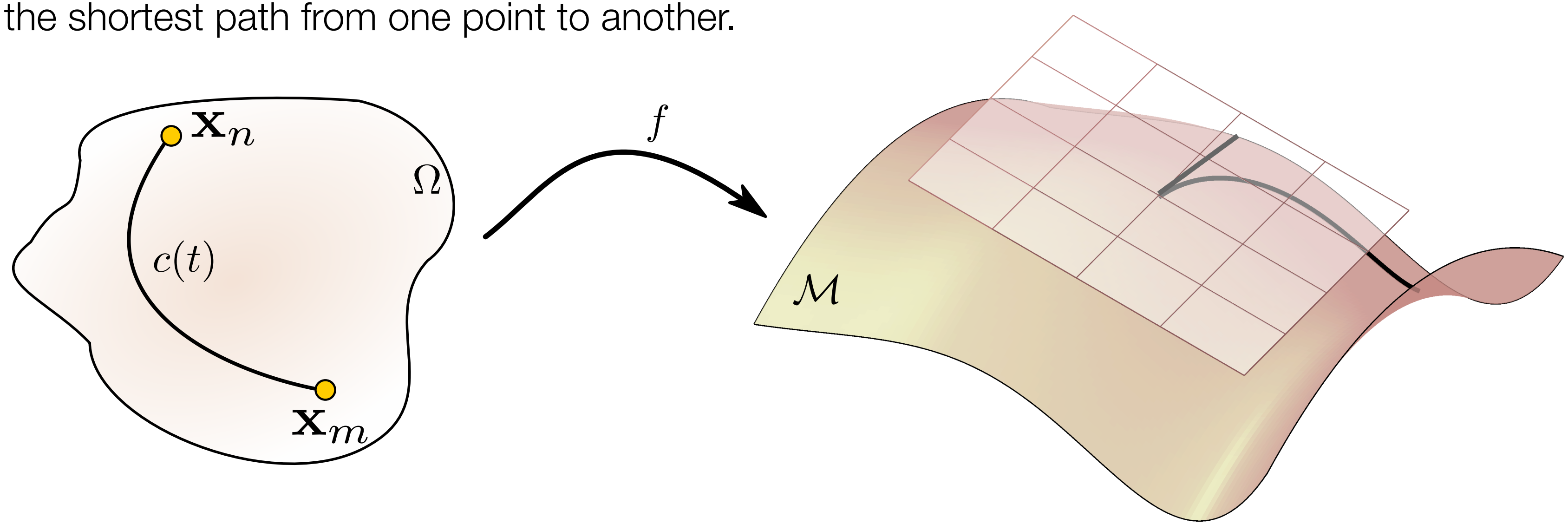

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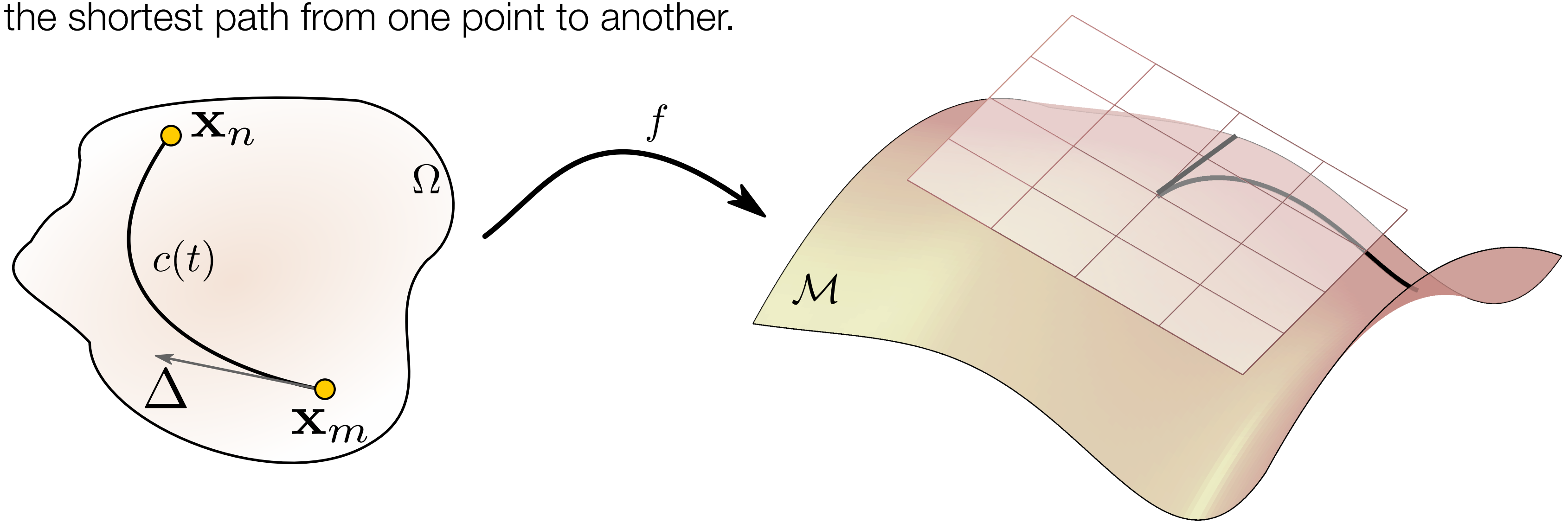

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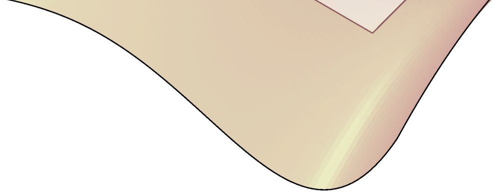
We, thus, define Riemannian subtraction as the initial velocity of the shortest path connecting two points. For historical reasons, this is called the *logarithm map*,

$$\mathrm{Log}_{\mathbf{x}_m}(\mathbf{x}_n) = \Delta = \partial_t c_{mn}(0)$$

(note this vector belongs to the tangent space of the initial point)



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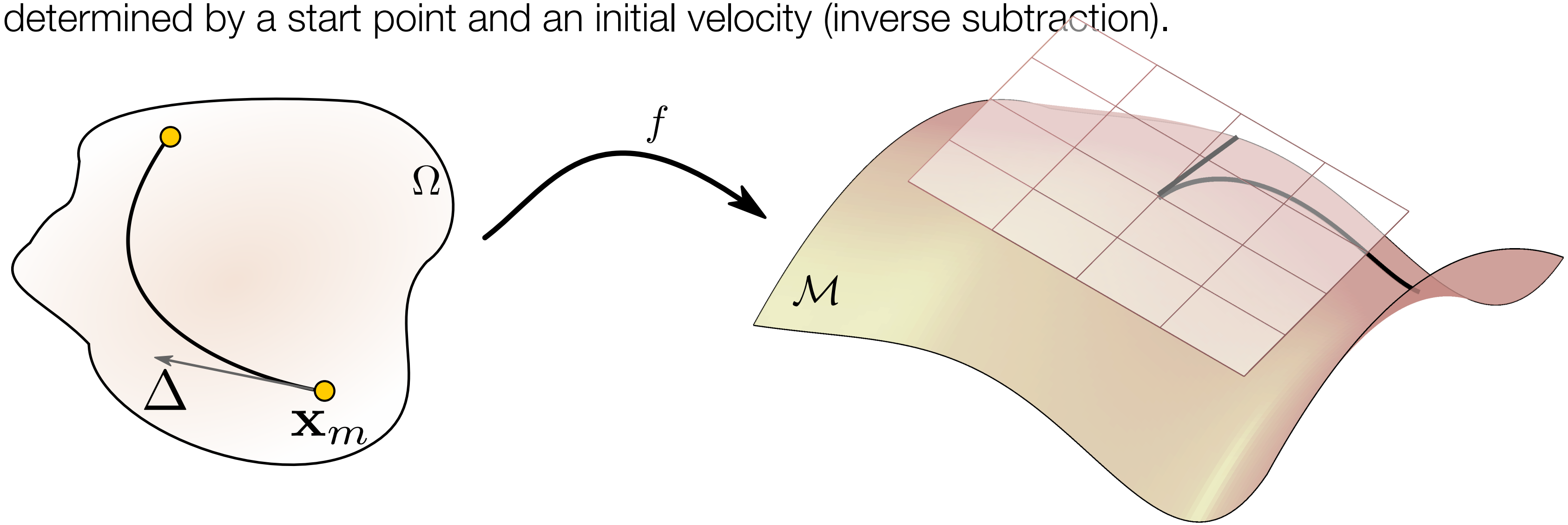
Since the geodesic is constant-speed, then we have

$$\begin{aligned}\mathrm{Length}\,(c_{mn}) &= \int_0^1 \|\partial_t c_{mn}(t)\| \mathrm{d}t \\ &= \|\partial_t c_{mn}(0)\| \int_0^1 \mathrm{d}t \\ &= \|\Delta\|.\end{aligned}$$

That is, the length of the vector Δ matches the length of the geodesic (just like Euclidean subtraction).

- Riemannian addition

Remember, addition is an operator which return a point along a shortest path determined by a start point and an initial velocity (inverse subtraction).



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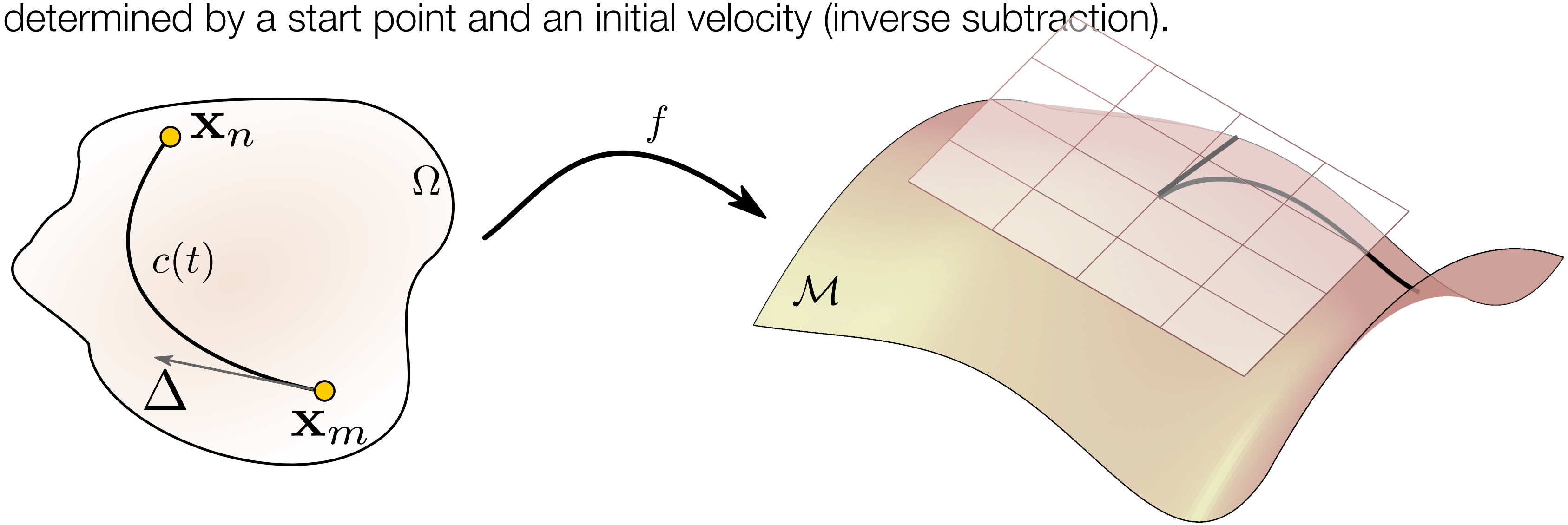


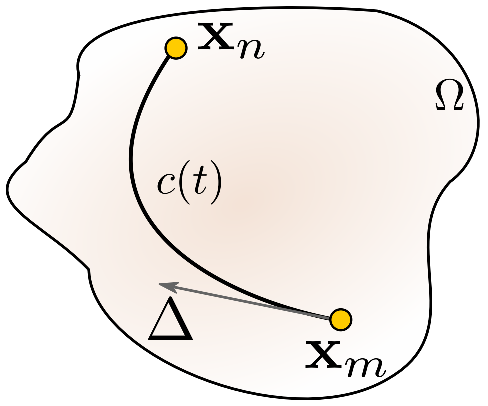
We know that the starting point together with the initial velocity of the geodesic determines the entire shortest path.

We, thus, define Riemannian addition as the end-point of the geodesic determined by a starting point and an initial velocity

$$\mathrm{Exp}_{\mathbf{x}_m}(\Delta) = \mathbf{x}_n.$$

~ Intermediate summary ~

We can generalize addition and subtraction by noting that geodesics are solutions to a system of 2nd order ODEs. Then we can get similar operators



Subtraction:

$$\mathrm{Log}_{\mathbf{x}_m}(\mathbf{x}_n) = \Delta = \partial_t c_{mn}(0)$$

Addition:

$$\mathrm{Exp}_{\mathbf{x}_m}(\Delta) = \mathbf{x}_n.$$

Exercises

You should complete exercises in the exercise sheet

Exam relevant

Exercise 6.1, 6.2, 6.3, 6.4
You should at least do one of these before moving on to the programming.

Don't be surprised if you see something like this at an exam...

Project relevant

Exercise 6.5, and 6.6:
Compute geodesics.
I strongly encourage you to do both exercises after having read them both
(much code can be shared).